

Thermiapt: Sensory Perception of Quantitative Thermodynamics Concepts in Education

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Figure 1: (A) *Thermiapt* is a haptic device for experiencing thermodynamics. (B) The device can present a range of materials.

Abstract

Traditional text-based methods for learning thermodynamics are often inefficient according to neuroscientific principles. Multi-sensory inputs, particularly visual and haptic, have been shown to enhance memory retention and comprehension by engaging the hippocampus. This paper presents *Thermiapt*, a haptic device designed to integrate sensory experiences of quantitative thermodynamic concepts, such as heat capacity. The device is not only simple and reproducible but also has the potential to significantly enhance understanding and retention in educational settings by leveraging multi-sensory learning.

Keywords

Education; Haptics; Thermal sensation; Virtual Reality

ACM Reference Format:

Anji Fujiwara, Kodai Iwasaki, Tamami Watanabe, and Hideaki Uchiyama. 2024. Thermiapt: Sensory Perception of Quantitative Thermodynamics Concepts in Education. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3681756.3697966>

1 Introduction

Learning thermodynamics involves transitioning from concrete phenomena to abstract concepts, which can be particularly challenging. It deals with macroscopic properties such as temperature and pressure, induced by the microscopic behavior of particles, making these concepts abstract and difficult for students to grasp [Cox et al. 2003].

Traditional learning methods, such as "plug-and-chug," rely heavily on passive memorization and simple substitution into formulas. These methods do not effectively foster a deep conceptual understanding, often leading to poor exam performance. Prior research has addressed this with visualization techniques to link microscopic and macroscopic properties [Cox et al. 2003; Hernández Bravo et al. 2020].

However, these methods remain limited to visual input, which presents a challenge in thermodynamics. While human sensory

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697966>

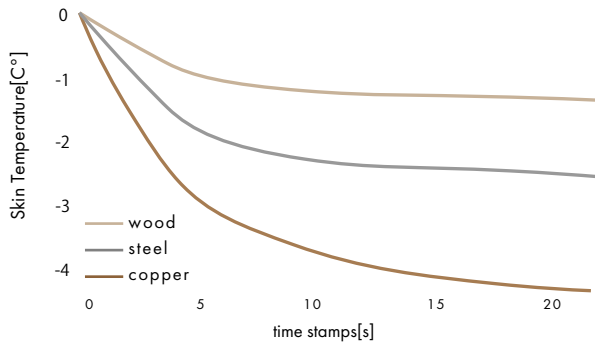


Figure 2: Temperature drop by material when the same amount of heat is applied.

input is predominantly visual, thermodynamics mainly involves heat and matter transport phenomena, which are best perceived through tactile input. This asymmetry between the sensory input and the nature of the subject matter hinders effective learning.

This paper presents *Thermiapt*, a haptic device that integrates sensory experiences of quantitative thermodynamic concepts through visual and haptic modalities. The primary objective is to improve understanding and retention of abstract thermodynamic concepts through a multi-sensory approach, which engages multiple senses to strengthen neural connections and facilitate better integration and retrieval of complex information [Chen 2020]. By providing haptic feedback that allows students to feel differences in heat capacity, our device helps them grasp complex principles often difficult to understand through visual input alone. Its simplicity and reproducibility also enhance its practicality for diverse educational settings.

2 Heat Capacity Perception

Heat capacity C is defined as:

$$C = \Delta Q / \Delta T \quad (1)$$

where ΔQ is the heat supplied, and ΔT is the temperature increase. A system with larger heat capacity experiences a smaller temperature increase for the same heat, while one with smaller heat capacity shows a larger increase (see Fig. 2).

When a finger supplies heat (Δq) to a material, the resulting temperature change (ΔT) is indicative of the material's heat capacity. Materials with higher heat capacity, such as wood, demonstrate smaller temperature changes, while those with lower heat capacity, like copper, exhibit larger changes. This difference facilitates tactile perception of varying heat capacities.

3 System Configuration

3.1 Hardware

To effectively represent quantitative concepts such as heat capacity and thermal conductivity in thermodynamics, it is essential to design a device capable of presenting various material sensations through a head-mounted display (HMD), specifically the Meta Quest 3, for educational purposes. We have developed such a device using

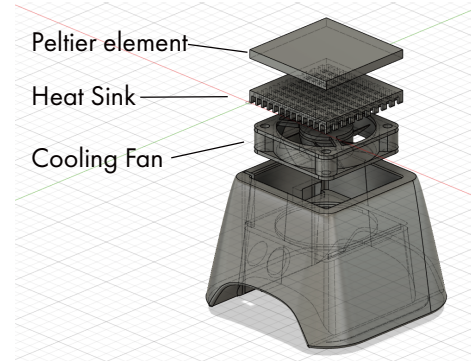


Figure 3: Hardware configuration of *Thermiapt*.

Peltier elements and an HMD to deliver both visual and haptic feedback. The system includes an ESP32 microcontroller, a temperature sensor (103JT-025), and a Peltier element (TETC1-12706-T100-SS-TF01-ALO) with a fan and heatsink to regulate the thermal output (see Fig. 3). This configuration simulates the transient thermal response of a human finger when contacting real-world materials, replicating the temperature change via the Peltier element.

3.2 Software

To accurately represent the relationship between the current applied to the Peltier element and the perceived temperature, we developed a calibration process. This process accounts for individual differences in temperature perception. The relationship is mathematically expressed as:

$$\Delta T = f(P_c) \quad (2)$$

where the temperature change ΔT is a function f of the current P_c applied to the Peltier element. Calibration of f was performed empirically by adjusting the coefficients based on individual responses to thermal stimuli. This ensures consistent temperature perception across all users.

Additionally, the positions of the real-world device and the virtual materials in the VR space were meticulously calibrated. Users can select different materials via the user interface, which adjusts the current to simulate the thermal properties of the selected material. This approach transitions from traditional text-based education to a potentially more efficient, experience-focused learning method.

4 Conclusion

This study demonstrates the potential of *Thermiapt* to enhance understanding and retention of thermodynamic concepts through sensory integration. By connecting theoretical knowledge with practical application, the system offers a more immersive learning environment. Its modular design ensures seamless curriculum integration, supporting tailored educational experiences. Future research will focus on controlled experiments to assess effectiveness, with the goal of validating and broadening its application in engineering education.

References

- Lihan Chen. 2020. Education and visual neuroscience: A mini-review. *PsyCh Journal* 9, 4 (2020), 524–532.
- Anne J Cox, Mario Belloni, Melissa Dancy, and Wolfgang Christian. 2003. Teaching thermodynamics with Physlets® in introductory physics. *Physics Education* 38, 5 (2003), 433.
- Martha Patricia Hernández Bravo, Adolfo Eduardo Obaya Valdivia, Carlos Montaña Osornio, and Yolanda Marina Vargas-Rodríguez. 2020. Didactic Strategy: Interactive Digital Board in Teaching Learning Heat Capacity for High School. *Online Submission* 9, 10 (2020), 216–223.